MASTER_PROJECT_SUMMARY_TPRM (Big File)

1. Project Title

Third-Party Risk Management (TPRM): Vendor Risk Analysis & Fraud Intelligence System

2. Project Overview

This project focuses on analyzing vendor risk and detecting potential fraud patterns within a Third Party Risk Management (TPRM) framework using SQL.

Organizations rely heavily on external vendors, which introduces risks related to cybersecurity, financial stability, and regulatory compliance. This project aims to ensure data quality, evaluate vendor risk exposure, and identify suspicious activities that may indicate fraud.

Through structured data analysis and automation techniques, the project shows how data driven insights can support better decision making in vendor management.

3. Objectives

The key objectives of this project include:

- Ensuring data accuracy and consistency through data cleaning
- Identifying high-risk vendors based on multiple risk dimensions
- Detecting fraud patterns and suspicious vendor behavior
- Analyzing vendor risk distribution across categories
- Developing automated reporting using stored procedures

4. Dataset Description

The dataset consists of two primary tables:

tprm_vendors

Contains vendor related information such as:

- Vendor Name
- Category
- Country
- Contract Value
- Risk Status
- Onboarding Date

vendor_audit

Contains audit and risk related data including:

- Cyber Security Score

- Financial Score
- Compliance Score
- Overall Risk Classification
- Audit Date

These tables are linked using a common **vendor_id**, enabling integrated analysis of vendor risk and performance.

5. Data Cleaning & Data Quality Assurance

Data quality is critical for accurate risk analysis. Several checks and transformations were performed:

- **Missing Vendor Records:** Identified full audit records without corresponding vendor entries to ensure referential integrity.
- **Invalid Risk Scores:** Detected scores outside the acceptable range (0–100), which could distort analysis results.
- **Missing Data:** Identified null values in critical fields such as audit dates and compliance scores.
- **Duplicate Records:** Detected 47 duplicate audit entries that could lead to inflated risk assessments.
- **Data Standardization:** Standardized category values and trimmed unnecessary spaces to ensure consistency.
- **Date Validation:** Identified future audit dates, which indicate data entry errors.

These steps ensured that the dataset is clean, reliable, and suitable for downstream analysis.

6. Risk Analysis

The project performs comprehensive risk analysis to evaluate vendor exposure:

- **Risk Distribution Analysis:** Assessed the proportion of vendors across high, medium, and low risk categories to understand overall portfolio exposure.
- **Weighted Risk Scoring:** Developed a scoring model combining cyber (50%), financial (30%), and compliance (20%) metrics to rank vendors.
- **Industry Level Risk Assessment:** Analyzed average risk scores across vendor categories to identify high-risk sectors.
- **High Risk Vendor Identification:** Flagged vendors with low scores and critical risk classifications.
- **Remediation Planning:** Generated actionable recommendations for vendors requiring intervention.
- The vendor risk portfolio shows a relatively balanced distribution across risk levels.
  Low-risk vendors account for 28%, indicating a stable portion of the vendor base.
  However, **Critical-risk vendors represent 25%**, which is significant and requires immediate attention.
  Additionally, **Medium-risk vendors make up 25%**, while **High-risk vendors account for 22%** of the total portfolio.
  This distribution highlights that nearly half of the vendors fall within high to critical risk categories, suggesting a considerable level of exposure that requires proactive monitoring and mitigation strategies. Combined, **47% of vendors fall into high and critical risk categories**, indicating elevated operational and security risks.

These analyses provide insights into where the organization is most vulnerable and where resources should be focused.

7. Fraud Intelligence & Anomaly Analysis

Fraud detection is a key component of vendor risk management. The following anomaly detection techniques were applied:

The analysis identified several patterns that may indicate potential fraud risks within the vendor dataset.

- **Suspicious Contract Values:**
  A total of **544 vendors** were identified with unusually high contract values exceeding expected thresholds. This represents a significant portion of the vendor base and may indicate potential overbilling, inflated contracts, or weak procurement controls.
- **Ghost Vendors (Missing Country):**
  No vendors (**0 cases**) were found with missing country information, suggesting strong data completeness and proper vendor onboarding processes in this area.
- **Onboarding Anomalies:**
  1 onboarding date showed an unusually high number of vendor registrations, indicating a potential bulk onboarding event that may require further investigation to rule out process irregularities.
- **Risk vs. Incident Mismatch:**
  No vendors (**0 cases**) were found to have low risk classification alongside high incident counts, suggesting consistency between risk assessment and incident tracking mechanisms.
- **Round Number Fraud Pattern:**
  A total of **11 vendors** exhibited round-number contract values, a known pattern associated with potentially manipulated or fabricated financial transactions.
- **Suspicious Contract Values:** Identified vendors with unusually high contract amounts, indicating potential overbilling.
- **Ghost Vendors:** Detected vendors with missing or undefined country data, which may signal incomplete or fraudulent records.
- **Onboarding Anomalies:** Identified unusual spikes in vendor onboarding activity that may require further validation.
- **Risk vs Incident Mismatch:** Flagged vendors classified as low risk but with high incident counts, indicating possible misclassification.
- **Round-Value Transactions:** Detected contract values in clean round numbers, a common pattern in fraudulent invoices.

These insights highlight potential fraud risks and improve organizational vigilance.

8. Stored Procedures & Automation

To enhance efficiency and scalability, stored procedures were developed:

- **Industry Based Risk Report:** Enables dynamic filtering of vendor risk by category for targeted analysis.
- **Top Vendor Identification:** Retrieves high-performing vendors based on defined score thresholds.
- **Compliance Alert System:** Automatically flags vendors with low compliance scores.

These procedures demonstrate how SQL can be used to automate reporting and support real-time decision-making in risk management systems.

9. Key Insights

- A significant portion of vendors fall into medium to high risk categories, indicating notable exposure.
- Certain vendor categories show consistently lower cyber security scores, increasing vulnerability to cyber threats.
- Multiple anomalies detected suggest potential fraud risks in contract values and onboarding processes.
- Data inconsistencies highlight the importance of strong data governance practices.
- Automated procedures can significantly improve efficiency in monitoring vendor risk.

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10. Conclusion & Recommendations

This project demonstrates the importance of integrating data quality, risk analysis, and fraud detection in vendor management.

Recommendations:

- Prioritize high-risk vendors for immediate review and remediation
- Strengthen vendor onboarding and validation processes
- Implement continuous monitoring systems for fraud detection
- Standardize data entry processes to reduce inconsistencies
- Leverage automated reporting tools for real time insights

By adopting these strategies, organizations can reduce operational, financial, and regulatory risks associated with third party vendors.